

Supporting Information

Post-Transition State Bifurcations in Reactions that Form Pimarenyl Cation Diastereomers – Implications for Terpene Biosynthesis

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1. Computational Details and Energy Table

Density functional theory (DFT) calculations were carried out using *Gaussian16 C.01*.¹ All geometries were optimized at the B3LYP-D3(BJ)/6-31+G(d,p) level of theory, which includes dispersion correction.²⁻⁷ Intrinsic reaction coordinate (IRC) calculations were used to confirm the minima connected to transition structures.⁸⁻¹⁰ Benchmarking was performed at the DLPNO-CCSD(T)/def2-TZVPP level of theory.¹¹⁻¹³ Grimme's quasi-harmonic thermochemical corrections were calculated with the *Goodvibes* package.¹⁴ Images of optimized structures were drawn with the CYLview2.0 software package.¹⁵

Data set collection of optimized geometries is available at the ioChem-BD repository¹⁶ via: <https://iochem-bd.bsc.es/browse/review-collection/100/478645/0d00175ed2134579ed9be110>.

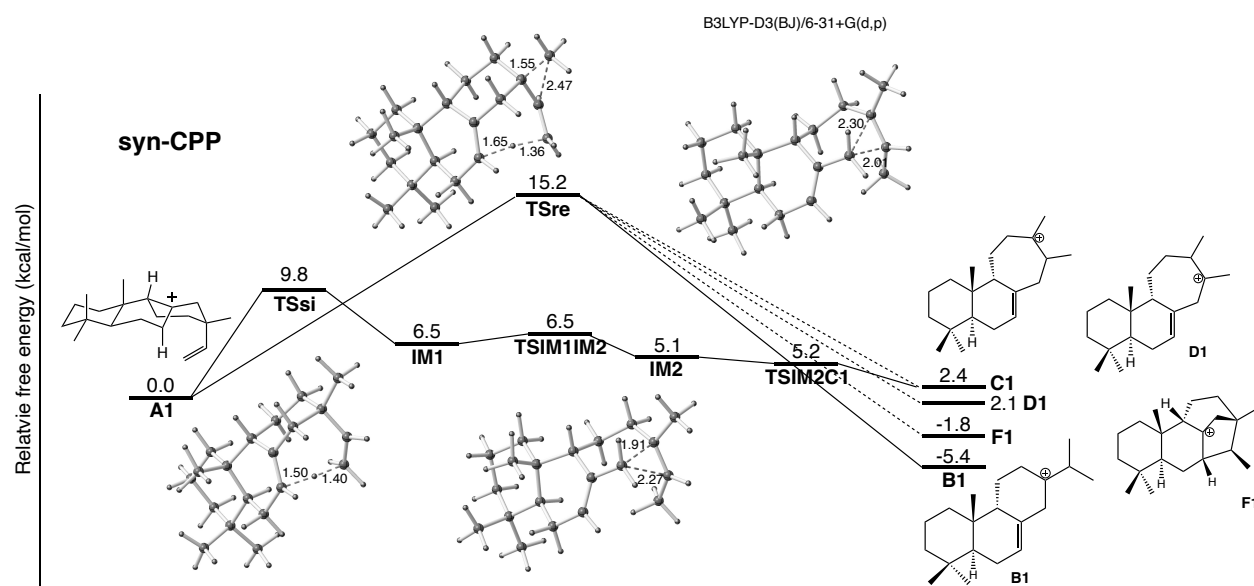


Figure S1. Complete energy profile for the syn-CPP system. Solid lines represent pathways connected by IRC, whereas dashed lines represent products from MD simulations.

Table S1. Electronic energies and free energy corrections for syn-CPP system structures shown in Figure S1 (units in Hartree).

Structure	Electronic Energy	Gibbs Free Energy Total	EE(DLPNO-CCSD(T)/def2-TZVPP)
A1_from TSre IRC	-781.908406	-781.463285	-780.2905381
TSre	-781.882567	-781.440209	-780.2638984
B1	-781.919608	-781.473203	-780.3008225
D1	-781.904755	-781.461127	-780.2838528
F1	-781.91729	-781.467448	-780.3058951

A1_from TSsi IRC	-781.91039	-781.464537	-780.2927101
TSsi	-781.8916	-781.448800	-780.2723817
IM1	-781.89889	-781.454233	-780.2785618
TSIM1IM2	-781.89869	-781.45417	-780.2789654
IM2	-781.90021	-781.45639	-780.2767016
TSIM2C1	-781.89989	-781.45618	-780.2785907
C1	-781.90529	-781.46067	-780.2849592

Table S2. Comparison of relative electronic energies for the computational accuracy assessment.

Structure	EE (B3LYP-D3(BJ)/6-31+G(d,p))	Relative EE (kcal/mol)	EE(DLPNO-CCSD(T)/def2-TZVPP)	Relative EE (kcal/mol)
A1_from TSre IRC	-781.908406	0	-780.2905381	0
TSre	-781.882567	16.2142051	-780.2638984	16.7166536
B1	-781.919608	-7.0293558	-780.3008225	-6.4535764
D1	-781.904755	2.29103536	-780.2838528	4.1950742
F1	-781.91729	-5.5754175	-780.3058951	-9.6366286
A1_from TSsi IRC	-781.91039	0	-780.2927101	0
TSsi	-781.8916	11.7871291	-780.2723817	12.7562453
IM1	-781.89889	7.2163535	-780.2785618	8.87819489
TSIM1IM2	-781.89869	7.34060028	-780.2789654	8.62491447
IM2	-781.90021	6.38302155	-780.2767016	10.0454441
TSIM2C1	-781.89989	6.58696197	-780.2785907	8.86007337
C1	-781.90529	3.19590334	-780.2849592	4.86378069

Table S3. Electronic energies and free energy corrections for *ent*-CPP system (unit in Hartree).

Structure	Electronic Energy	Gibbs Free Energy Total
A1_from TSre IRC	-781.908314	-781.462417
TSre	-781.882567	-781.440209
B1	-781.880782	-781.438393
A1_from TSsi IRC	-781.908849	-781.462292
TSsi	-781.885649	-781.442076
C1	-781.89878	-781.453206

Table S4. Electronic energies and free energy corrections for related proton transfer system (unit in Hartree).

Structure	Electronic Energy	Gibbs Free Energy Total
A3_fromA3C3irc	-781.912217	-781.464961
TSA3C3	-781.880206	-781.438411
B3	-781.916068	-781.470294
C3	-781.901028	-781.455479
D3	-781.904073	-781.458273
E3	-781.938306	-781.494162
F3	-781.913009	-781.464375
A2	-781.902404	-781.457936
A3_from A3D2irc	-781.91192	-781.465038
TSA3D2	-781.884717	-781.443294
B2	-781.922422	-781.477113
C2	-781.895942	-781.452770
D2	-781.90105	-781.456351
E2	-781.938306	-781.494162
F2	-781.917138	-781.468840

2. AIMD Simulation Details

Quasi-classical downhill dynamics were propagated with Verlet integrator using the *progdyn* code developed by the Singleton group.¹⁷ Trajectories were initiated in both the reactant and product directions from the region of a particular transition state structure. The timestep was 1 fs, and trajectories were terminated at 500 fs if no product was formed. “Excluded trajectories” in the following table include reactant recrossing, product recrossing, different-product recrossing, and different-reactant recrossing cases.

a. Stopping criteria for each structure and product distribution.

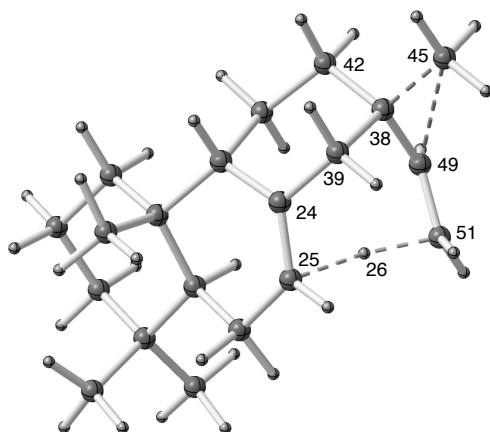


Figure S2. Atom labels used for defining coordinates in AIMD simulation for syn-CPP.

Table S4. Bond distance stopping criteria for each structure in the syn-CPP AIMD simulations.

Traj Initiated	Product Formation	Stopping Criteria (unit in Å)	# of trajectories
syn-CPP TSre	Reactant	C25H26<1.2	121
	B1 product	(H26C51<1.2) & (C45C49<1.6)	105
	C1 product	(H26C51<1.2) & (C39C49<1.6)	5
	D1 product	(H26C51<1.2) & (C42C49<1.6)	11
	Excluded trajectories		33
syn-CPP TSsi	Reactant	C25H26<1.2	96
	B1 product	(H26C51<1.2) & (C45C49<1.6)	0
	C1 product	(H26C51<1.2) & (C39C49<1.6)	96
	D1 product	(H26C51<1.2) & (C42C49<1.6)	0
	Excluded trajectories		14

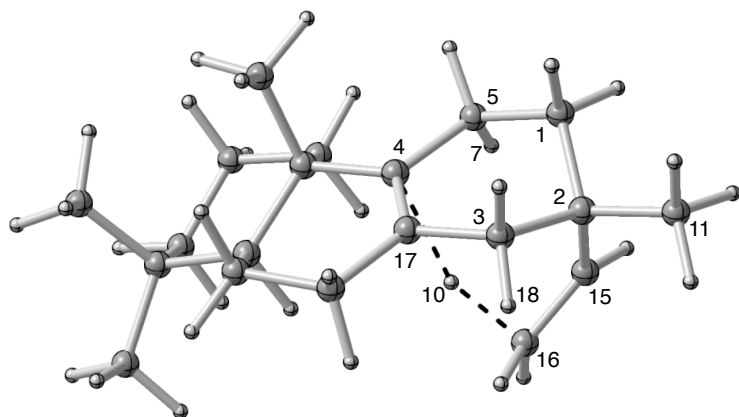


Figure S3. Atom labels used for defining coordinates in AIMD simulation for CPP.

Table S5. Bond distance stopping criteria for each structure in the CPP AIMD simulations (Level of theory: B3LYP-D3(BJ)/6-31g(d)).

Traj Initiated	Product Formation	Stopping Criteria (unit in Å)	# of trajectory
CPP TSsi	Reactant1	C4H10<1.2	122
	Reactant 2	C17H10<1.2	32
	B1 product	(C16H10<1.2) & (C11C15<1.6)	1
	C1 product	(C16H10<1.2) & (C3C15<1.6)	73
	D1 product	(C16H10<1.2) & (C1C15<1.6)	5
	2 nd carbocation		59
	F1 product	(C16H10<1.2) & (C15H18<1.9)	1
	I1 product	(C16H10<1.2) & (C15H7<1.2)	1
	L1 product	(C16H10<1.2) & (C2C3>2.3)	14
	Excluded trajectories		20
CPP TSre	Reactant1	C4H10<1.2	61
	Reactant 2	C17H10<1.2	81
	B1 product	(C16H10<1.2) & (C11C15<1.6)	12
	C1 product	(C16H10<1.2) & (C3C15<1.6)	5
	D1 product	(C16H10<1.2) & (C1C15<1.6)	78
	2 nd carbocation		28
	F1 product	(C16H10<1.2) & (C15H18<1.9)	10
	J1 product	(C16H10<1.2) & (C15C17<1.9)	6
	N1 product	(C16H10<1.2) & (C1C5>2.3)	1
	K1 product	(C16H10<1.2) & (C4C15<1.6)	2
	Excluded trajectories		20

Table S6. Bond distance stopping criteria for each structure in the CPP AIMD simulations (Level of theory: B3LYP/6-31+g(d,p)).

Traj Initiated	Product Formation	Stopping Criteria (unit in Å)	# of trajectory
CPP TSsi	Reactant1	C4H10<1.2	47
	Reactant 2	C17H10<1.2	9
	B1 product	(C16H10<1.2) & (C11C15<1.6)	0
	C1 product	(C16H10<1.2) & (C3C15<1.6)	33
	D1 product	(C16H10<1.2) & (C1C15<1.6)	0
	2 nd carbocation		15
	I1 product	(C16H10<1.2) & (C15H7<1.2)	1
	L1 product	(C16H10<1.2) & (C2C3>2.3)	7
	Excluded trajectories		3
CPP TSre	Reactant1	C4H10<1.2	9
	Reactant 2	C17H10<1.2	52
	B1 product	(C16H10<1.2) & (C11C15<1.6)	21
	C1 product	(C16H10<1.2) & (C3C15<1.6)	4
	D1 product	(C16H10<1.2) & (C1C15<1.6)	24
	2 nd carbocation		7
	F1 product	(C16H10<1.2) & (C15H18<1.9)	2
	J1 product	(C16H10<1.2) & (C15C17<1.9)	2
	N1 product	(C16H10<1.2) & (C1C5>2.3)	0
	K1 product	(C16H10<1.2) & (C4C15<1.6)	1
	Excluded trajectories		14

Table S6. Bond distance stopping criteria for each structure in the CPP AIMD simulations (Level of theory: B3LYP-D3(BJ)/6-31g(d)).

Traj Initiated	Product Formation	Stopping Criteria (unit in Å)	# of trajectory
CPP TSsi	Reactant1	C4H10<1.2	36
	Reactant 2	C17H10<1.2	10
	B1 product	(C16H10<1.2) & (C11C15<1.6)	0
	C1 product	(C16H10<1.2) & (C3C15<1.6)	18
	D1 product	(C16H10<1.2) & (C1C15<1.6)	1
	2 nd carbocation		20
	N1 product	(C16H10<1.2) & (C1C5>2.3)	3
	I1 product	(C16H10<1.2) & (C15H7<1.2)	3
	L1 product	(C16H10<1.2) & (C2C3>2.3)	1
	Excluded trajectories		10
CPP TSre	Reactant1	C4H10<1.2	24
	Reactant 2	C17H10<1.2	30
	B1 product	(C16H10<1.2) & (C11C15<1.6)	10
	C1 product	(C16H10<1.2) & (C3C15<1.6)	2
	D1 product	(C16H10<1.2) & (C1C15<1.6)	28
	2 nd carbocation		7

	F1 product	(C16H10<1.2) & (C15H18<1.9)	3
	I1 product	(C16H10<1.2) & (C15H7<1.2)	1
	J1 product	(C16H10<1.2) & (C15C17<1.9)	3
	Excluded trajectories		17

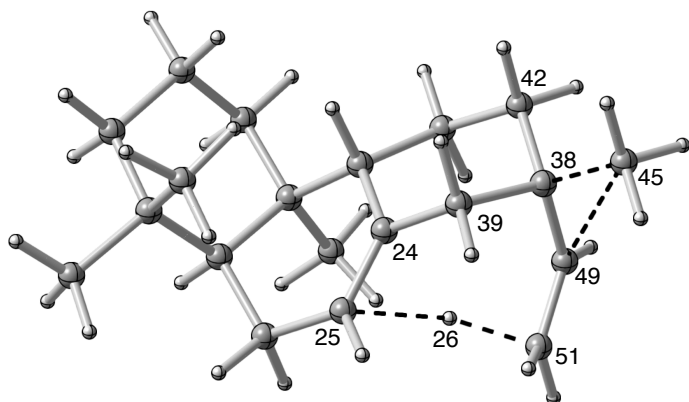


Figure S4. Atom labels used for defining coordinates in AIMD simulation for ent-CPP.

Table S6. Bond distance stopping criteria for each structure in the ent-CPP AIMD simulations.

Traj Initiated	Product Formation	Stopping Criteria (unit in Å)	# of trajectories
ent-CPP TSsi	A1 reactant	C25H26<1.2	106
	B1 product	(H26C51<1.2) & (C45C49<1.6)	0
	C1 product	(H26C51<1.2) & (C39C49<1.6)	106
	D1 product	(H26C51<1.2) & (C42C49<1.6)	0
	Excluded trajectories		22
ent-CPP TSre	A1 reactant	C25H26<1.2	110
	B1 product	(H26C51<1.2) & (C45C49<1.6)	94
	C1 product	(H26C51<1.2) & (C39C49<1.6)	3
	D1 product	(H26C51<1.2) & (C42C49<1.6)	10
	2 nd carbocation		3
	Excluded trajectories		28

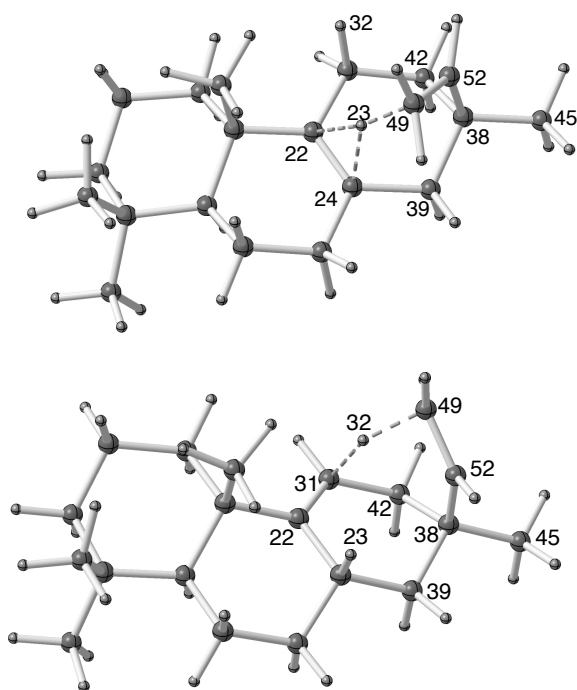


Figure S5. Atom labels used for defining coordinates in AIMD simulation for A3D2 (top) and A3C3 (bottom).

Table S7. Bond distance stopping criteria for each structure in the related proton transfer AIMD simulations.

Traj Initiated	Product Formation	Stopping Criteria (unit in Å)	# of trajectories
A3D2	A2 reactant	C22H23<1.2	67
	A3 reactant	H23C24<1.2	11
	B2 product	(H23C49<1.2) & (C45C52<1.6)	18
	C2 product	(H23C49<1.2) & (C39C52<1.6)	5
	D2 product	(H23C49<1.2) & (C42C52<1.6)	42
	E2 product	(H23C49<1.2) & (H32C52<1.2)	3
	F2 product	(H23C49<1.2) & (C24C52<1.6)	9
	Excluded trajectories		5
A3C3	A2 reactant	C22H32<1.2	0
	A3 reactant	H32C31<1.2	59
	B3 product	(H32C49<1.2) & (C45C52<1.6)	22
	C3 product	(H32C49<1.2) & (C39C52<1.6)	33
	D3 product	(H32C49<1.2) & (C42C52<1.6)	1
	E3 product	(H32C49<1.2) & (H23C52<1.2)	1
	F3 product	(H32C49<1.2) & (C31C52<1.6)	2
	Excluded trajectories		5

b. Progdyn Setup

Progdyn.conf

#This is the configuration file for PROGDYN. This file is read by progdynstarterHP and
the awk programs proggenHP, prog1stpoint, prog2ndpoint, and progdynb.
#The programs won't read anything past the first blank line,
#and this file must end with a blank line.
#The program has a number of default values but they are unlikely to be what you want.
#Do not delete lines - rather, comment out lines for unwanted options.
#The values here are read repeatedly and most can be changed in the middle of running
jobs
****The keywords are case sensitive. The following keywords should always be defined:****
****method, charge, multiplicity, memory, processors, title
**** method --The following word is copied exactly to the gaussian input file.
method b3lyp/6-31g(d)
#To do a nonstandard route, make nonstandard 1. For normal calcs, use nonstandard 0 or
else leave it out.
#Then make a file called "nonstandard" containing the nonstandard route with no extra
lines.
#nonstandard 0
NMRoptions NMRtype 1 will add a section for an NMR calc at every NMRevery intervals.
The NMRmethod is the calculation used
#for the NMR calculation. The awk program progNMRm might be useful as a starting point
for pulling out the data from files named
#NMRlist
#NMRtype 1
#NMRmethod ub3lyp/genecp
#NMRevery 4
#Use geometry linear to get the program to recognize 3N-5 normal modes instead of 3N-6
when reading from freqinHP
#geometry linear
#Rotationmode controls whether molecular rotations are turned on. They should be turned
on with rotationmode 1 but this can be set
#at 0 to reproduce older calculations.
#rotationmode 1
**** method2 --The options here are restricted, unrestricted, and read. restricted is the
default
#If the method is U..., put unrestricted here and the .com files will have in them guess=mix.
#If you put read here, the .com files will contain guess=tcheck, which sometimes makes
things faster, sometimes not.
#The use of read requires a specifically defined checkpoint file name using the keyword
checkpoint.

method2 restricted
charge 1
multiplicity 1
processors 32
#*** memory --The following "word" is copied exactly to the gaussian input file
after %mem=.
memory 64GB
#*** killcheck and checkpoint -- You can use a specifically defined checkpoint file name by
putting
#the name after the keyword checkpoint. This is necessary if you use the read option with
method2.
#Defined checkpoint names are an unnecessary modest hassle and if you do not want to
bother, use killcheck 1
killcheck 1
#checkpoint g09.chk
#*** diagnostics -- 0 prints out nothing extra, 1 (default) prints out extra stuff to a
#file "diagnostics", 2 adds more stuff, 3 adds velocities to a file "vellist"
#4 adds the apparent temperature to vellist, but this is meaningless with quasiclassical
calculations
diagnostics 2
#*** title -- the title keyword must be followed by exactly four words
title cpp si md test
#*** initialdis -- 0 (default) turns off displacement of the normal modes, so that all
trajectories start from the same place
and only the energies and signs of the motion in the modes are randomized
1 gives a flat distribution of displacements where all of the possible values are equally
likely
2 (recommended) gives a QM-like gaussian distribution of displacements, so that
displacements in the middle are more likely than
those at the end by 1/e
initialdis 2
#*** timestep -- this is the time between points in the trajectory. Typical values would be
1E-15 or 0.5E-15 or 0.25E-15
timestep 1E-15
#*** scaling -- this lets you scale the gaussian frequencies by a constant
scaling 1.0
temperature 298.15
#*** method3, method4, method5, and method6 -- These keywords let you add extra lines
to the gaussian input file.
#method3 and method4 add lines at the top of the input after the lines defining the
method, and
#this is useful to implement things like the iop for mPW1k
#method5 and method6 add lines after the geometry, after a blank line of course

```

#only a single term with no spaces can be added, one per method line. Here are some
examples to uncomment if needed
method3 empiricaldispersion=gd3bj
#method3 SCRF=(SMD,solvent=DMSO)
#add the line below with big structures to get it to put out the distance matrix and the input
orientation
method4 iop(2/9=2000)
#method4 scrf=(pcm,solvent=dmsol,read)
#method4 IOp(3/76=1000001970)IOp(3/77=0800008000)IOp(3/78=0700010000)
#method5 radii=bondi
#method6
#*** methodfile -- This keyword lets you add more complicated endings to gaussian input
files
#such as a gen basis set. Put after the keyword the number of lines in a file you create
called
#methodfile that contains the test you want to add to the end of the gaussian input
methodfile 0
#*** numimag --This tells the program the number of imaginary frequencies in the starting
structure.
#if 0, treats as ground state and direction of all modes is random
#if 1, motion along the reaction coordinate will start out in the direction defined by
searchdir
#if 2, only lowest freq will go direction of searchdir and other imag mode will go in random
direction
numimag 1
#*** searchdir -- This keyword says what direction to follow the mode associated with the
imaginary frequency.
#The choices are "negative" and "positive". Positive moves in the direction defined in the
gaussian frequency calculation
#for the imaginary frequency, while negative moves in the opposite direction. The correct
choice can be made either
#by a careful inspection of the normal modes and standard orientation geometry, or by trial
and error.
searchdir negative
#*** classical -- for quassiclassical dynamics, the default, use 0. for classical dynamics,
use 1
#if there are no normal modes and the velocities are to be generated from scratch, use
classical 2
classical 0
#*** DRP, saddlepoint, and maxAtomMove --to run a DRP use 'DRP 1' in the line below,
otherwise leave it at 0 or comment it out
#the treatment of starting saddlepoints is not yet implemented so use saddlepoint no
#if DRP shows oscillations then decrease maxAtomMove
#DRP 1

```

```

#saddlepoint no
#maxAtomMove 0.01
*** cannonball -- The program can "fire" a trajectory from a starting position toward a
particular target, such as toward
#a ts. To use this, make a file cannontraj with numAtom lines and three numbers per line
that defines the vector
#for firing the trajectory, relative to the starting geometry's standard orientation. The
number following cannonball sets
#the extra energy being put into the structure in kcal/mol
cannonball 0
*** keepevery --This tells the program how often to write the gaussian output file to file
dyn, after the first two points.
#Use 1 for most dynamics to start with, but use a higher number to save on disk space or
molden loading time.
keepevery 1
*** highlevel --For ONIOM jobs, the following line states the number of highlevel atoms,
#which must come before the medium level atoms. Use some high value such as 999 if not
using ONIOM
highlevel 999
#linkatoms 1
*** fixedatom1, fixedatom2, fixedatom3, and fixedatom4 - These fix atoms in space.
#Fixing one atom serves no useful purpose and messes things up, while fixing two atoms
#fixes one distance and fixing three has the effect of fixing three distances, not just two
#in current form fixed atoms only are meant to work with no displacements, that is,
initialdis=0
#fixedatom1 2
#fixedatom2 3
#fixedatom3 19
*** boxon and boxsize - With boxon 1, a cubic box is set such that atoms that reach the
edge
#are reflected back toward the middle. Useful for dynamics with solvent molecules. This
is a crude
#implementation that is ok for a few thousand femtoseconds but will not conserve energy
long term.
#Set the box size so as to fit the entire initial molecule but not have too much extra room.
#The dimensions of the box are two times the boxsize, e.g. boxsize 7.5 leads to a box that is
15 x 15 x 15 angstroms
boxon 0
boxsize 7.5
*** displacements -- This keyword lets you set the initialdis of particular modes by using a
series of lines of the format
# displacements NumberOfMode InitialDisForThatMode, as in the example below. You
should be able to do as many of these as you like

```

you might consider this for rotations where a straight-line displacement goes wrong at large displacements

The choices for InitialDisForThatMode are 0, 1, 2, and 10, where 10 does the same thing as 0 but is maintained for now because

a previous version of the program had a bug that made 0 not work.

displacements 2 0

displacements 3 0

displacements 4 0

displacements 5 0

displacements 6 0

displacements 7 0

displacements 8 0

displacements 9 0

displacements 10 0

displacements 11 0

**** etolerance --This sets the allowable difference between the desired energy in a trajectory and the actual

#energy, known after point 1 from the potential energy + the kinetic energy in the initial velocities.

#The unit is kcal/mol and 1 is a normal value for mid-sized organic systems. For very large and floppy molecules, a larger value

#may be needed, but the value must stay way below the average thermal energy in the molecule (not counting zpe).

#If initialdis is not 0 and few trajectories are being rejected, decrease the value.

etolerance 1

**** controlphase --It is sometimes useful to set the phase of particular modes in the initialization of trajectories.

#The format is controlphase numberOfModeToControl positive or controlphase numberOfModeToControl negative.

#controlphase 3 positive

**** damping -- The damping keyword lets you add or subtract energy from the system at each point, by multiplying the velocities

#by the damping factor. A damping of 1 has no effect, and since you mostly want to change the energy slowly, normal values range

#from 0.95 to 1.05. The use of damping lets one do simulated annealing - you add energy until the structure is moving enough

#to sample the kinds of possibilities you are interested in, then you take away the energy slowly.

damping 1

**** reversetraj --This keyword sets the trajectories so that both directions from a transition state are explored.

reversetraj true

#updated Aug 9, 2007 to include the possibility of classical dynamics by the keyword classical

#updated Jan 2008 to include fixed atoms, ONIOM jobs, keepevery, and box size
#update Feb 2008 to include methodfile parameter
updated Nov 2008 to allow for start without an initial freq calc using classical = 2
update Aug 2010 to include etolerance, damping controlphase and reversetraj

Proganal (here, we use CPP si-face AIMD as example)

```
BEGIN {
firsttitle=1
getline < "isomernumber"
isomer=$1
}
/ cpp/ {
  if (firsttitle==1) {
    printf("%s %s %s %s %s %s %s %s ",$1,$2,$3,$4,$6,$7,$8)
    runpoint=$6
  }
  firsttitle++
}
/Standard orientation/,/Rotational constants/ {
  if (($1>.5) && ($1<99)) {
    A[$1]=$4;B[$1]=$5;C[$1]=$6
  }
}

END {
C25H26=Distance(25,26)
H26C51=Distance(26,51)
C39C49=Distance(39,49)
C45C49=Distance(45,49)

printf("%d %s %.3f %s %.3f %s %.3f %s %.3f",runpoint,"C25H26",C25H26,"H26C51",H26C51,
"C39C49",C39C49,"C45C49",C45C49)
  if (runpoint>500) {
    print "Too many points. XXXXN"
  }
  # system("date > nogo")
}
  if ((C25H26<1.2)) {
    print "Reactant formed XXXX"
  }
  if ((H26C51<1.2) && (C39C49<1.6)) {
    print "P1 formed XXXX"
  }
}
```

```

if ((H26C51<1.2) && (C45C49<1.6)) {
    print "P2 formed XXXX"
}
system("date '+%b:%d:%Y %T'")
system("tail -1 Echeck | grep XXXX")
}

```

```

function Distance(Atom1,Atom2) {
    return sqrt((A[Atom1]-A[Atom2])^2 + (B[Atom1]-B[Atom2])^2 + (C[Atom1]-C[Atom2])^2)
}

```

```

function Angle(Atom1,Atom2,Atom3) {
    value=((-
Distance(Atom1,Atom3)^2+Distance(Atom1,Atom2)^2+Distance(Atom2,Atom3)^2)/(2*Dis
tance(Atom1,Atom2)*Distance(Atom2,Atom3)))
    return acos(value)
}

```

```

function asin(x) { return (180/3.141592)*atan2(x, sqrt(1-x*x)) }

```

```

function acos(x) { return (180/3.141592)*atan2(sqrt(1-x*x), x) }

```

```

function atan(x) { return (180/3.141592)*atan2(x,1) }

```

```

function Dihedral(Atom1,Atom2,Atom3,Atom4) {
    B1x=A[Atom2]-A[Atom1]
    B1y=B[Atom2]-B[Atom1]
    B1z=C[Atom2]-C[Atom1]
    B2x=A[Atom3]-A[Atom2]
    B2y=B[Atom3]-B[Atom2]
    B2z=C[Atom3]-C[Atom2]
    B3x=A[Atom4]-A[Atom3]
    B3y=B[Atom4]-B[Atom3]
    B3z=C[Atom4]-C[Atom3]
    modB2=sqrt((B2x^2)+(B2y^2)+(B2z^2))
    # yAx is x-coord. etc of modulus of B2 times B1
    yAx=modB2*(B1x)
    yAy=modB2*(B1y)
    yAz=modB2*(B1z)
    # CP2 is the crossproduct of B2 and B3
    CP2x=(B2y*B3z)-(B2z*B3y)
    CP2y=(B2z*B3x)-(B2x*B3z)
    CP2z=(B2x*B3y)-(B2y*B3x)
    termY=((yAx*CP2x)+(yAy*CP2y)+(yAz*CP2z))
}

```

```

# CP is the crossproduct of B1 and B2
CPx=(B1y*B2z)-(B1z*B2y)
CPy=(B1z*B2x)-(B1x*B2z)
CPz=(B1x*B2y)-(B1y*B2x)
termX=((CPx*CP2x)+(CPy*CP2y)+(CPz*CP2z))
dihed4=(180/3.141592)*atan2(termY,termX)
return dihed4
}

function killdyn(isomer) {
  system("rm -f dyn")
}

```

3. Natural Bond Orbital (NBO) Analysis

Natural Bond Orbital (NBO) were calculated with NBO7.¹⁸ Since this is a concerted asynchronous reaction, when we apply NBO analysis, there are two interactions we are interested in. Before hydrogen transfer, the key interaction we are interested in is bonding orbital of the sigma bond we are interested in and the anti-bonding pi orbital on the vinyl group. After hydride transfer, the key interaction is still the bonding orbital of the sigma bond we are interested in but with the empty p orbital on the carbocation. The changing in each NBO value from reactant to transition state were summarized in Figure S6.

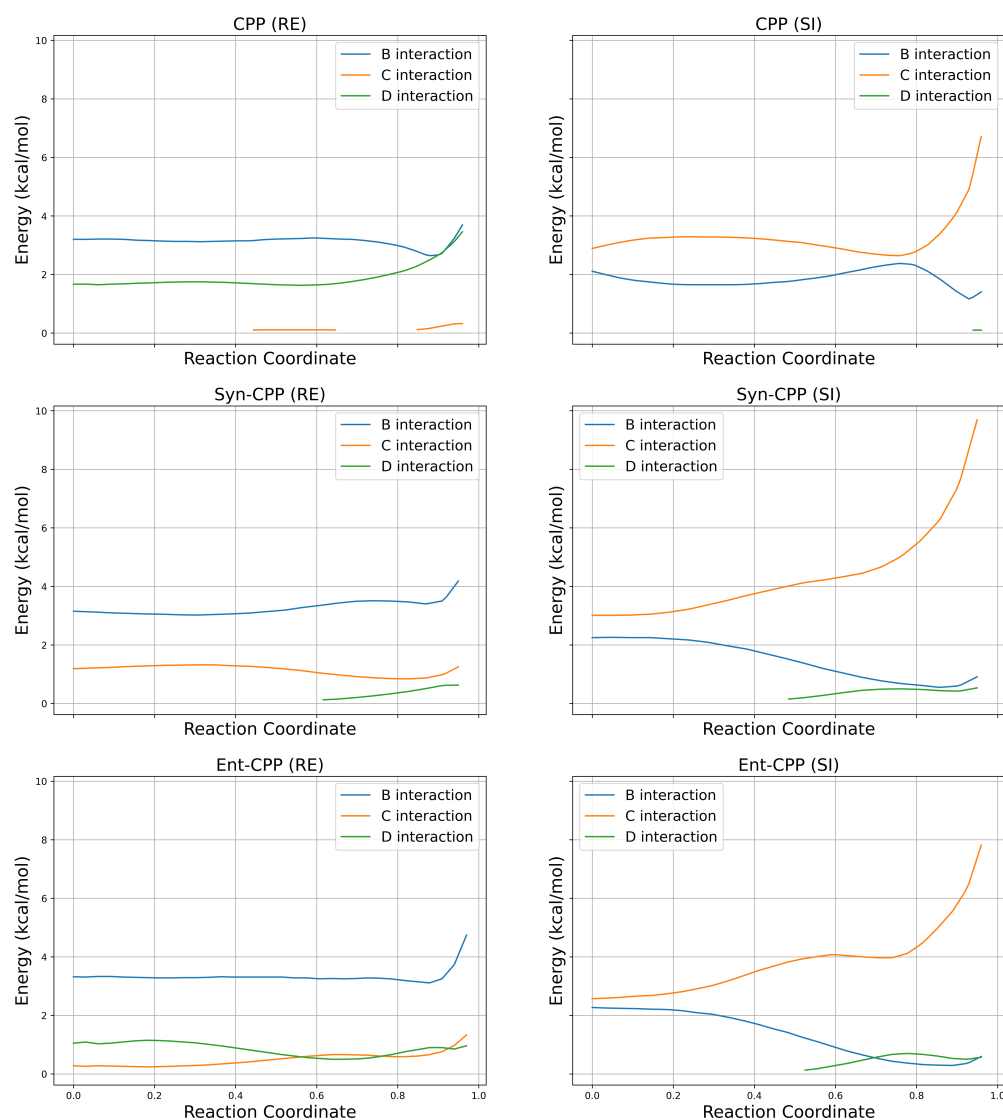


Figure S6. Changes in NBO interaction energies from reactant to transition state for CPP, syn-CPP, and ent-CPP systems. Each curve represents the key donor–acceptor orbital interaction associated with the formation of products B, C, and D, respectively.

Trajectory analysis indicates that re-face transition states generally lead to a higher proportion of bifurcated products, particularly in the syn- and ent-CPP systems. This observation is consistent with the NBO results. In si-face attack pathways, the orbital interaction associated with product C formation is more dominant than the interaction forming other products near the transition state, suggesting a reduced tendency for bifurcation. In contrast, for re-face attacks, the NBO interactions contributing to B, C, and D product formations are more evenly distributed. Although the B-forming interaction remains slightly stronger, its less dominant nature relative to the si-face case may contribute to the observed bifurcation in product outcomes.

We next examined how NBO interactions change after the transition state. Due to the distortion of orbital spaces as the system approaches product formation, the acceptor orbitals are no longer strictly unoccupied. This violates the assumptions of second-order perturbation theory, often resulting in artificially high interaction energies. To avoid this artifact, we restricted our analysis to the first 80 femtoseconds following the transition state, focusing on how orbital interaction tendencies evolve in trajectories that ultimately yield different products.

Because the Intrinsic Reaction Coordinate (IRC) pathway captures only the steepest descent toward a single product, we turned to molecular dynamics (MD) trajectory simulations to analyze post-transition state behavior. The results are summarized in Figure S7. Again, the re-face attack transition state exhibits greater product bifurcation, particularly in the syn-CPP and ent-CPP systems. Therefore, we focused our analysis on trajectories initiated from re-face transition states. Figure S3 illustrates the changes of NBO interaction energies along the trajectory for each system (CPP, syn-CPP, and ent-CPP). For each case, we randomly selected one trajectory that led to the formation of products B, C, and D, respectively. The plotted curves show how key donor–acceptor interactions change over time. In general, the donor–acceptor interaction associated with a given product tends to increase along the trajectory leading to its formation.

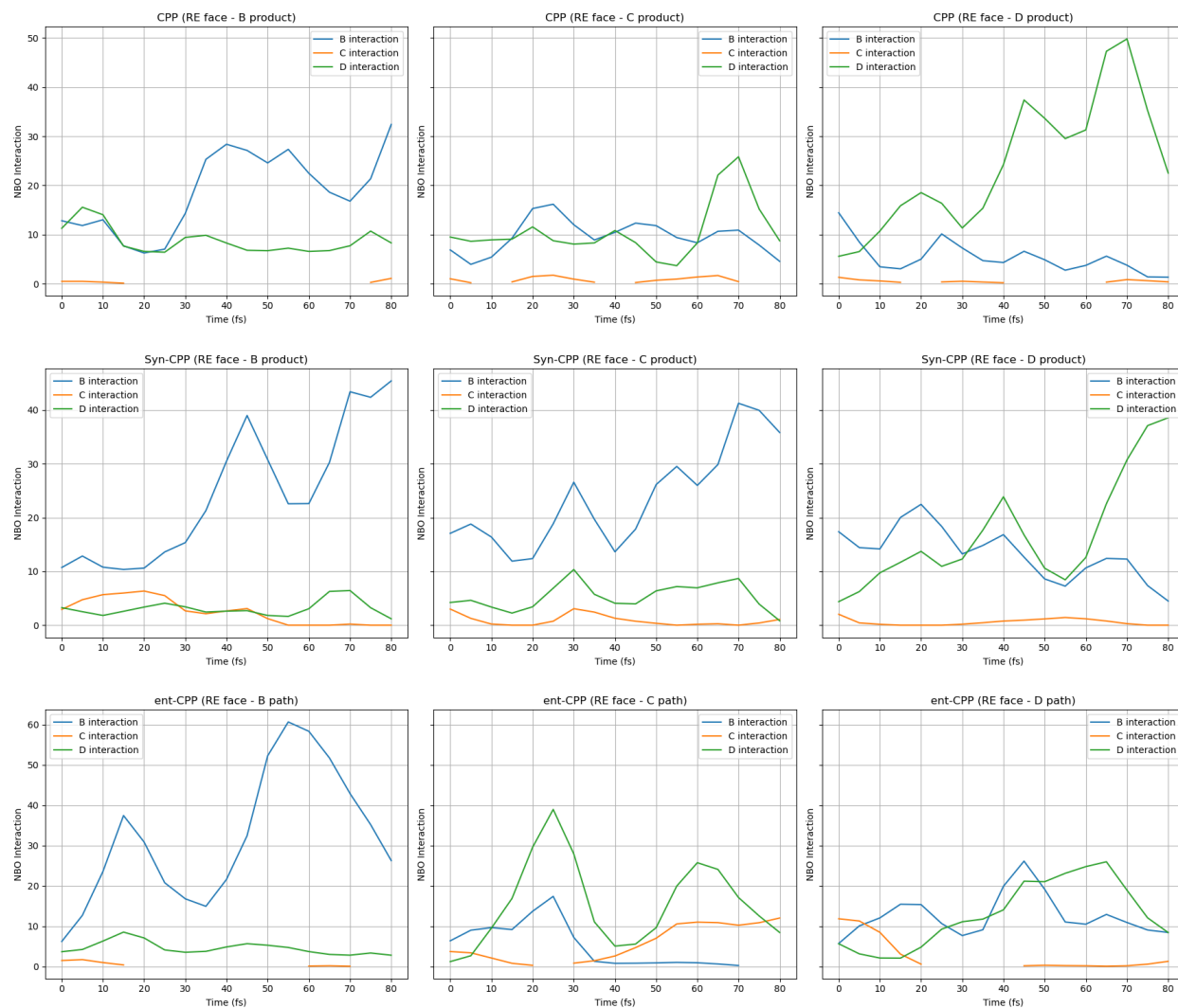


Figure S7. Time evolution of NBO interaction energies along post-transition state trajectories that ultimately lead to the formation of products B, C, or D. Each plot corresponds to a specific product-forming trajectory, with curves representing the key orbital interactions associated with B, C, and D product formation. Data are shown for CPP, syn-CPP and ent-CPP re-face attack systems.

4. Potential Energy Surface (PES) Construction Details and Ref. 14 Labels

Two principal components were generated using scikit-learn¹⁹ PCA based on the variance of key bond distances. These components were then used as the two axes of the potential energy surface.

Syn-CPP System Additional Plot

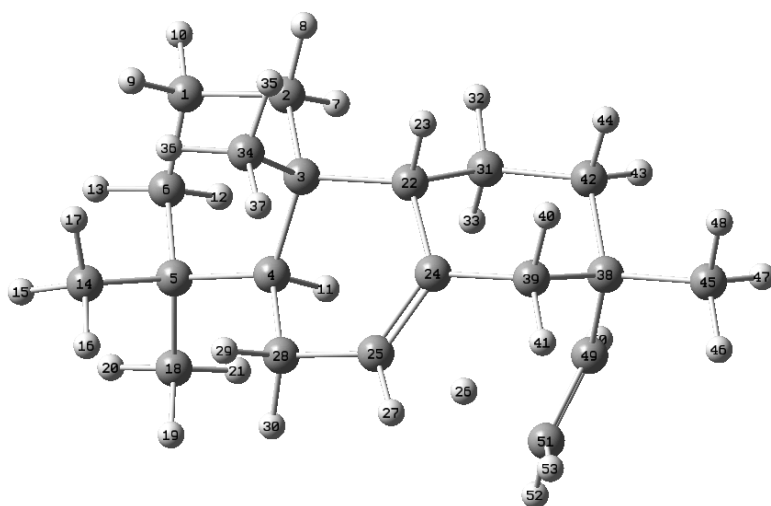


Figure S8. Atom labels used for defining coordinates in the potential energy surface for syn-CPP system. For example, an axis labeled R(1,2) in later figures indicates the distance between atoms 1 and 2.

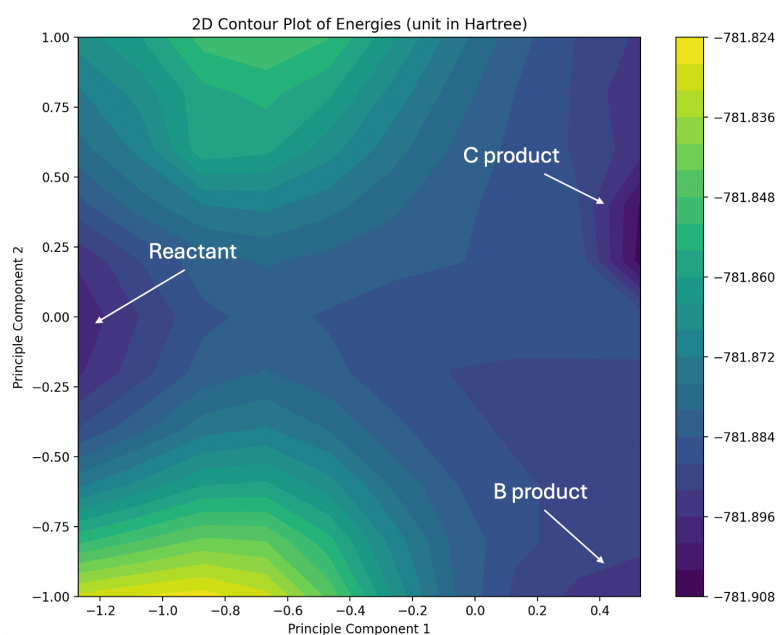


Figure S9. PES from reactant to either B or C product near Transition state for syn-CPP re system (top).
 $PC1 = 0.7208 \cdot (R(25,26) - 3.9670) - 0.5889 \cdot (R(25,51) - 2.9151) + 0.1739 \cdot (R(38,45) - 3.2830) - 0.2023 \cdot (R(45,49) - 4.2736) + 0.1402 \cdot (R(38,39) - 3.3302) - 0.2069 \cdot (R(39,49) - 4.4340)$
 $PC2 = 0.0206 \cdot (R(25,26) - 3.9670) + 0.0002 \cdot (R(25,51) - 2.9151) - 0.4920 \cdot (R(38,45) - 3.2830) + 0.5081 \cdot (R(45,49) - 4.2736) + 0.4970 \cdot (R(38,39) - 3.3302) - 0.5023 \cdot (R(39,49) - 4.4340)$

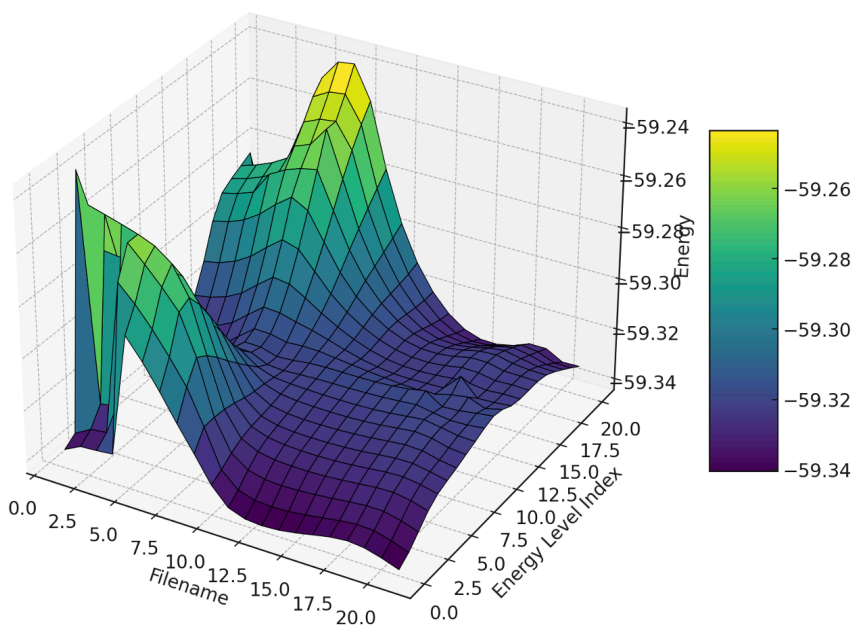


Figure S10. 3D version of plot Figure S5.

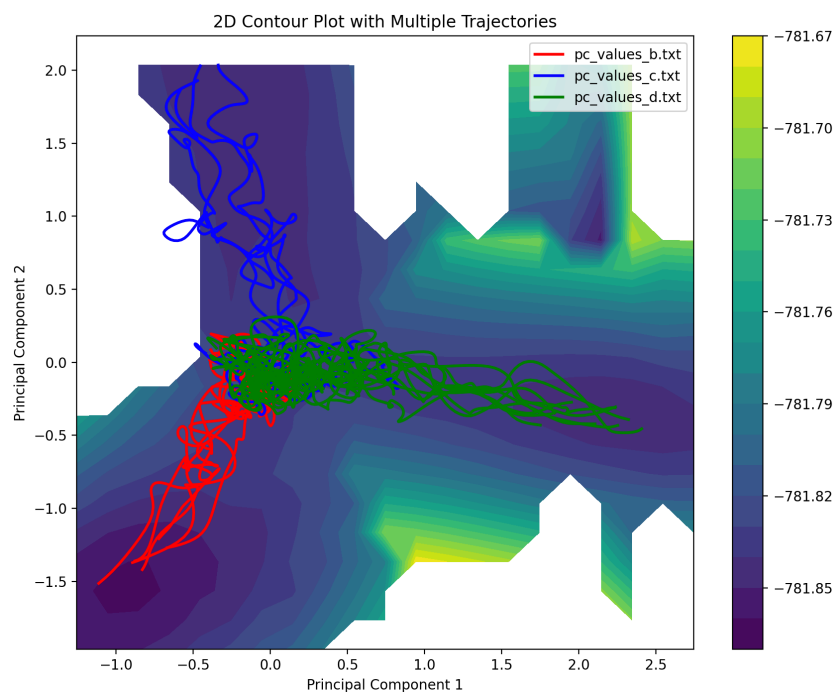


Figure S11. 2D potential energy surface with AIMD trajectories plotted on the surface.
 $(PC1=(0.1072)*(R(25,26)-2.5325)+(-0.021)*(R(26,51)-1.1575)+(-0.3403)*(R(38,45)-1.7225)+(0.3542)*(R(45,49)-2.2725)+(-0.1646)*(R(38,39)-1.7275)+(0.1547)*(R(39,49)-2.345)+(0.5922)*(R(38,42)-1.7425)+(-0.5874)*(R(42,49)-2.295))$
 $PC2=(0.0089)*(R(25,26)-2.5325)+(0.0023)*(R(26,51)-1.1575)+(-0.4342)*(R(38,45)-1.7225)+(0.4475)*(R(45,49)-2.2725)+(0.5388)*(R(38,39)-1.7275)+(-0.5429)*(R(39,49)-2.345)+(-0.1171)*(R(38,42)-1.7425)+(0.1109)*(R(42,49)-2.295))$

3D Surface Plot of Potential Energy Surface

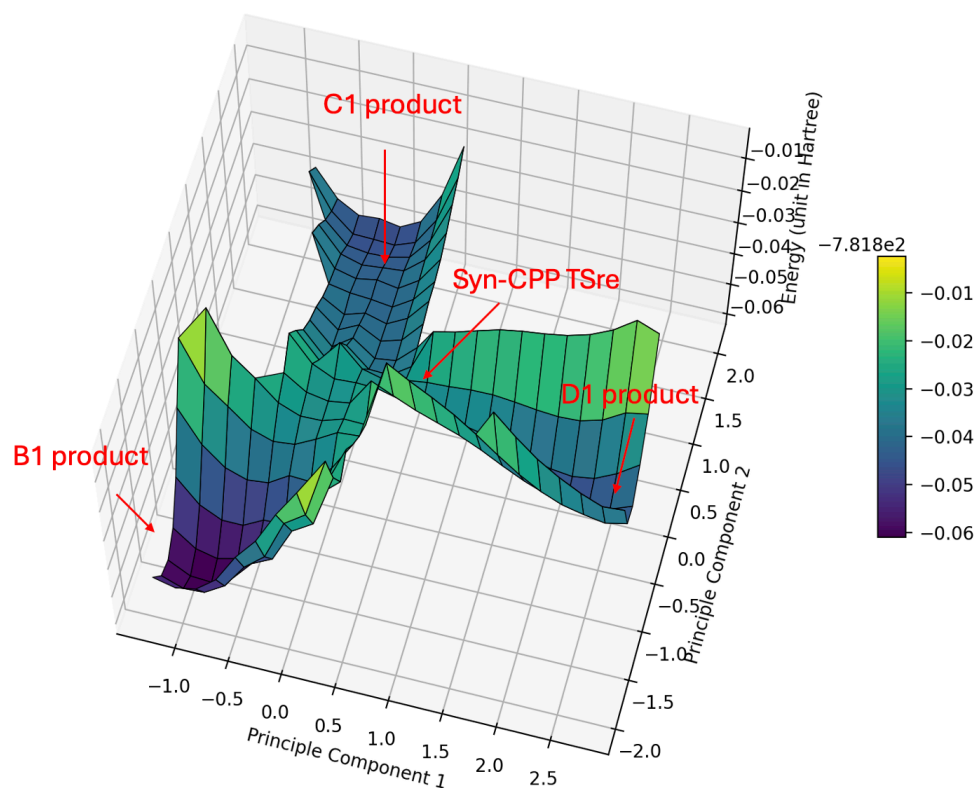


Figure S12. 3D plot of the syn-CPP re system showing the transition state and three products.

CPP system and Ref. 14 labels

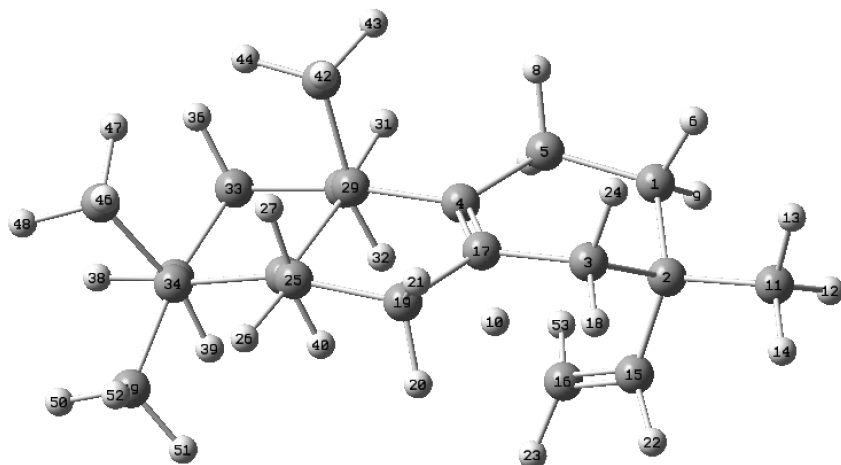


Figure S13. Atom labels used for defining coordinates in the potential energy surface for CPP system.

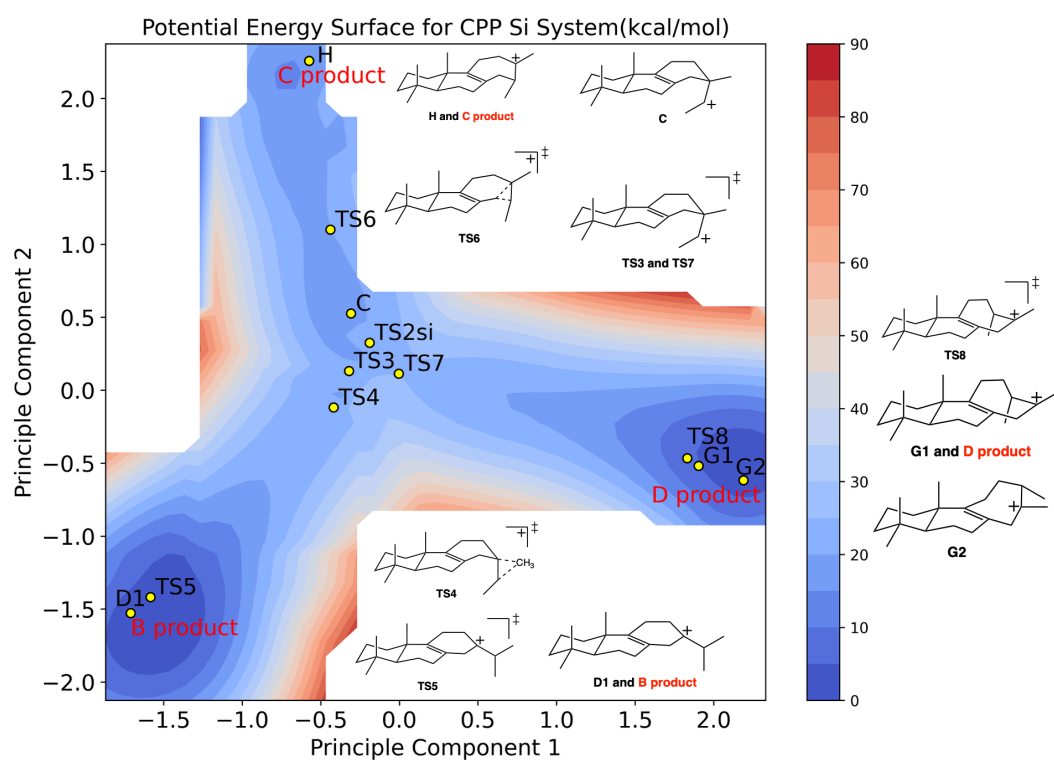


Figure S14. Structures of the saddle points from Ref. 14 corresponding to the CPP re system potential energy surface shown in Figure 5 ($PC1 = (-0.1218) \cdot (R(2,3) - 3.0886) + (0.1458) \cdot (R(3,15) - 4.4667) + (-0.3445) \cdot (R(2,11) - 3.1909) + (0.3513) \cdot (R(11,15) - 4.3801) + (0.5822) \cdot (R(1,2) - 3.2917) + (-0.6188) \cdot (R(1,15) - 4.3722)$ $PC2 = (0.4585) \cdot (R(2,3) - 3.0886) + (-0.5036) \cdot (R(3,15) - 4.4667) + (-0.4938) \cdot (R(2,11) - 3.1909) + (0.4809) \cdot (R(11,15) - 4.3801) + (-0.1661) \cdot (R(1,2) - 3.2917) + (0.1829) \cdot (R(1,15) - 4.3722)$).

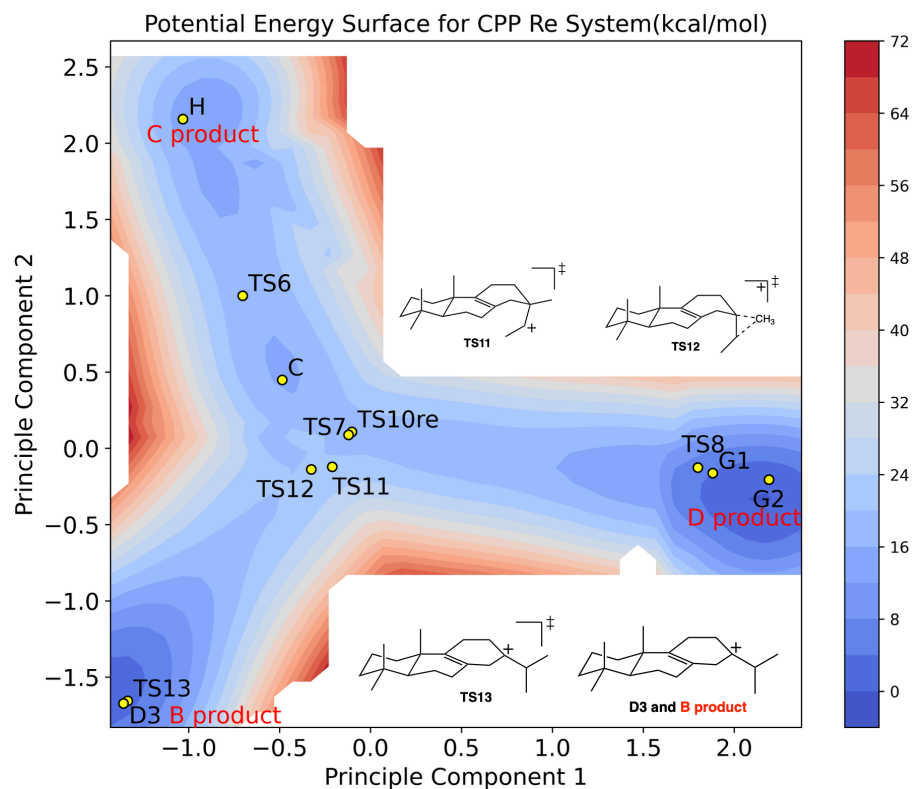


Figure S15. Structures of the saddle points from Ref. 14 corresponding to the CPP re system potential energy surface shown in Figure 5 ($PC = (-0.1494) \cdot (R(2,3) - 3.0697) + (0.167) \cdot (R(3,15) - 4.5092) + (-0.315) \cdot (R(2,11) - 3.1673) + (0.3298) \cdot (R(11,15) - 4.4021) + (0.5945) \cdot (R(1,2) - 3.3059) + (-0.6231) \cdot (R(1,15) - 4.3186)$ $PC2 = (0.469) \cdot (R(2,3) - 3.0697) + (-0.5237) \cdot (R(3,15) - 4.5092) + (-0.4829) \cdot (R(2,11) - 3.1673) + (0.4896) \cdot (R(11,15) - 4.4021) + (-0.1212) \cdot (R(1,2) - 3.3059) + (0.1348) \cdot (R(1,15) - 4.3186)$)).

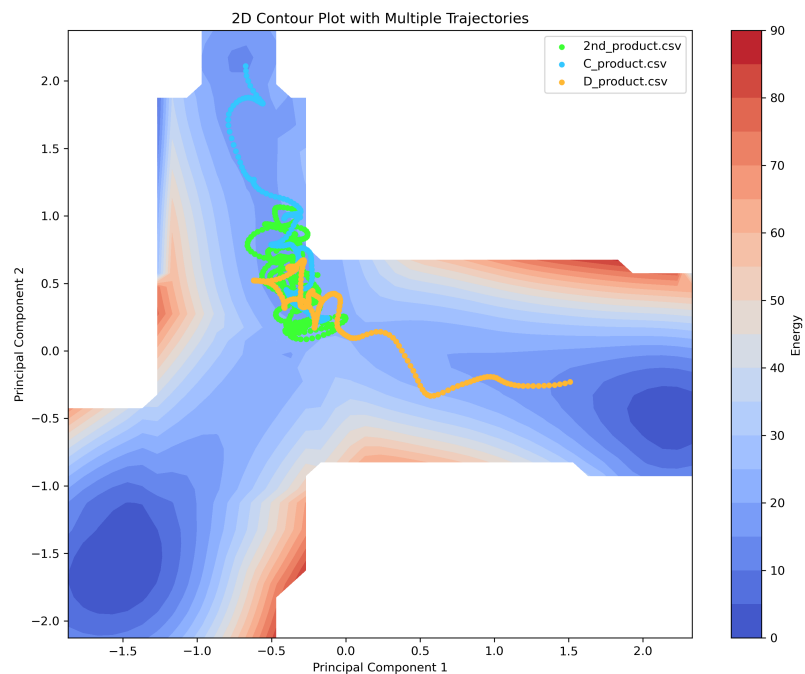


Figure S16. PES with AIMD trajectories plotted on the surface for CPP-si system.

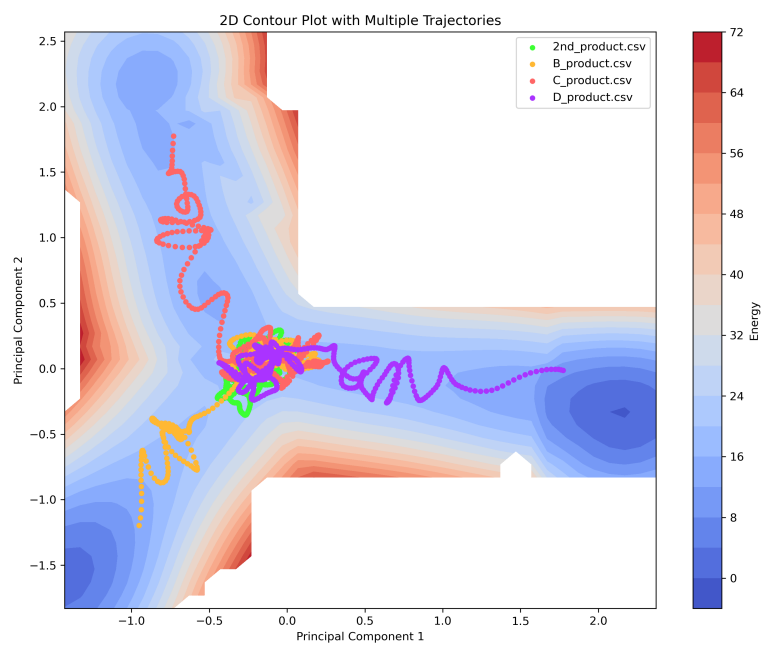


Figure S17. PES with AIMD trajectories plotted on the surface for CPP-re system.

5. Intrinsic Reaction Coordinate (IRC) Plots

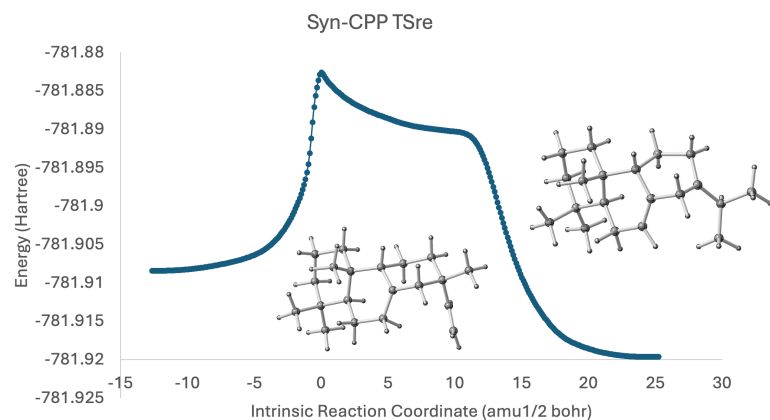


Figure S18. Syn-CPP TSre IRC.

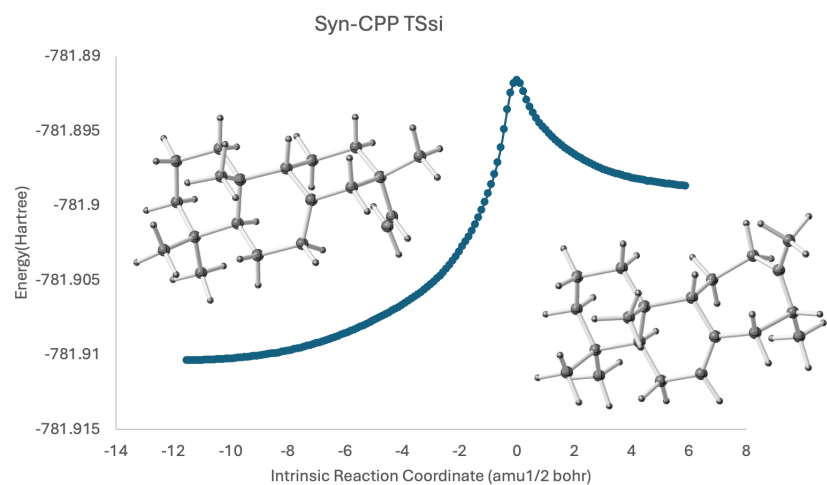


Figure S19. Syn-CPP TSsi IRC.

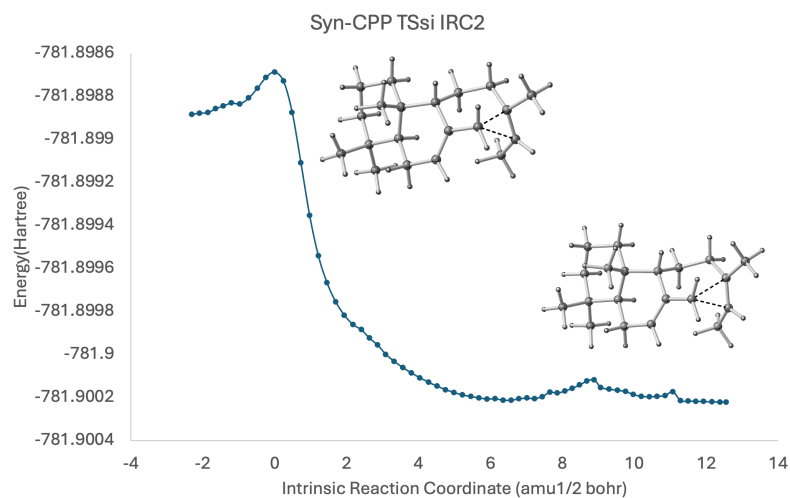


Figure S20. Syn-CPP TSsi IRC2.

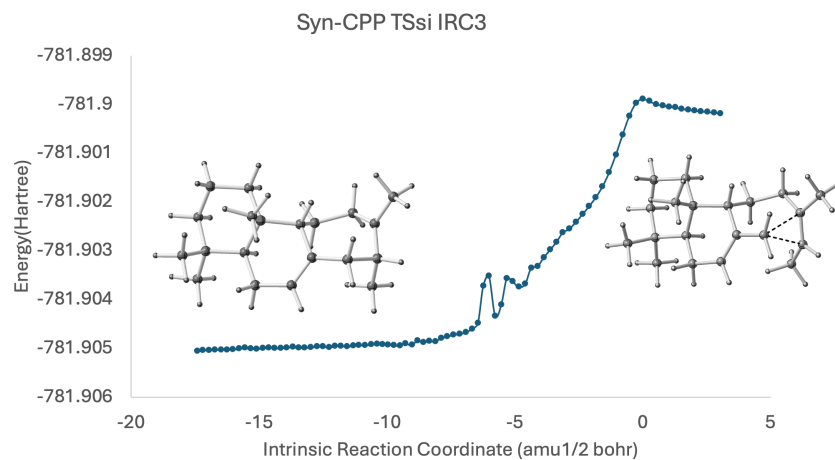


Figure S21. Syn-CPP TSsi IRC3.

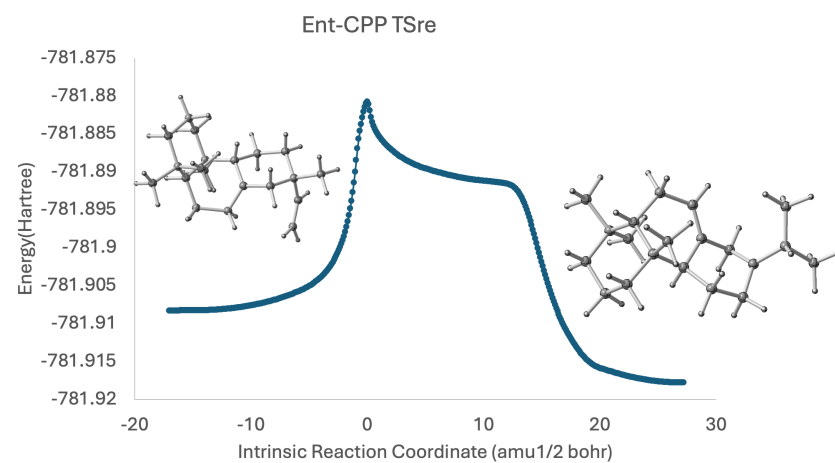


Figure S22. Ent-CPP TSre IRC.

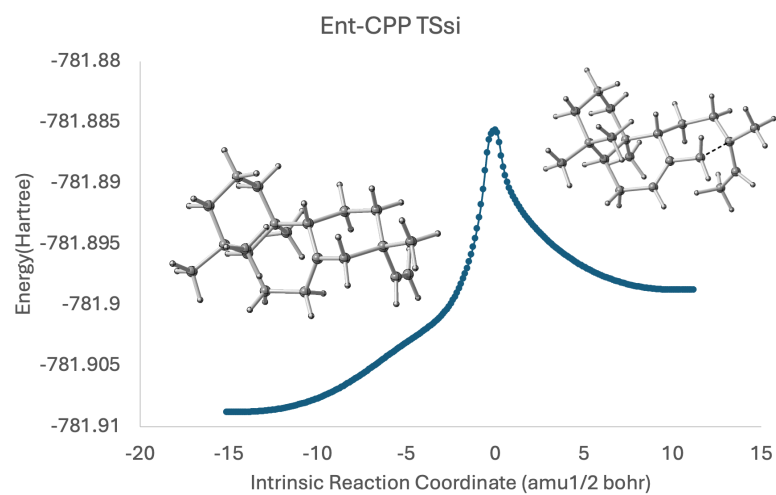


Figure S23. Ent-CPP TSsi IRC.

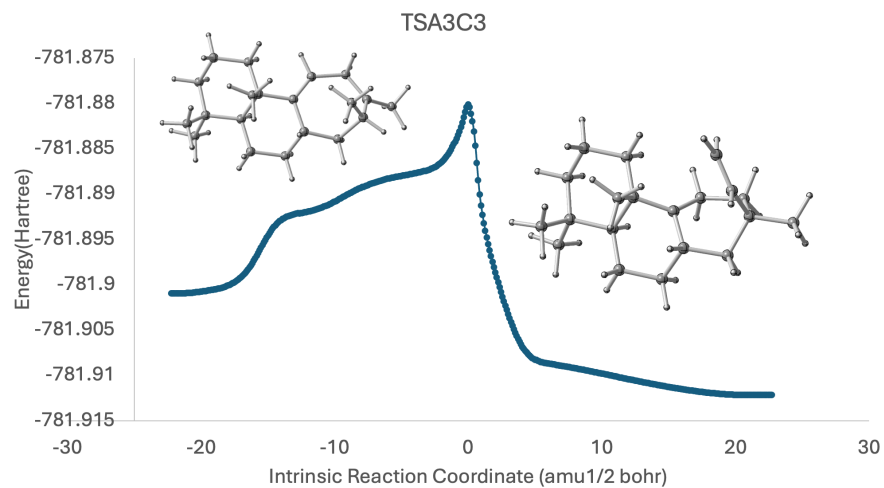


Figure S24. TSA3C3 IRC.

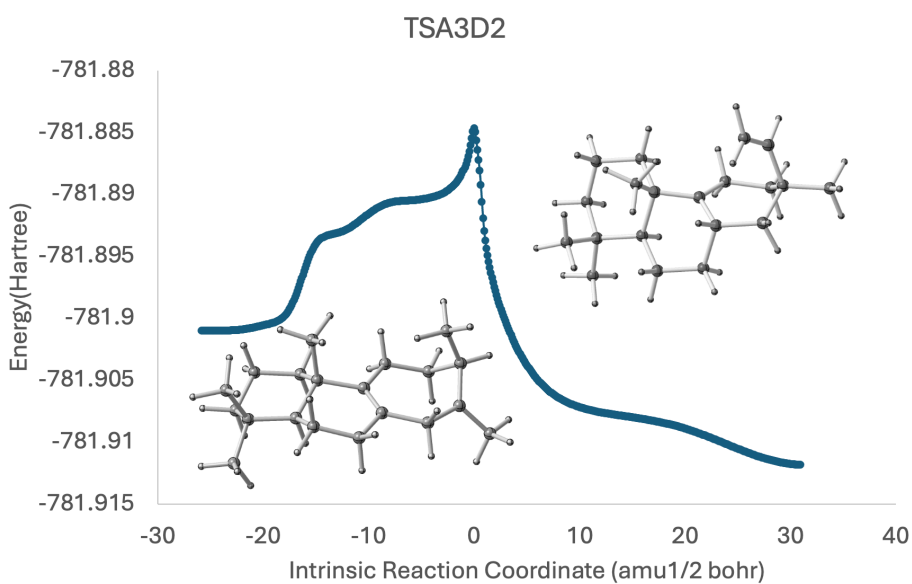


Figure S25. TSA3D2 IRC.

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